

Security Group vs Network Access Control List.

Security Group (SG):

What it is:

A **Security Group** is a **stateful firewall** attached to a resource (like EC2).

What “stateful” really means:

If you allow inbound traffic → **response is automatically allowed**.

You don't need to configure return traffic.

Key points:

- Works at **instance level**
- Only supports **ALLOW** rules
- Default: **deny everything**
- Very simple and safe

Example:

Allow HTTP:

- Inbound: port 80 → ALLOW
Done. No outbound rule needed.

Network ACL (NACL):

What it is:

A **NACL** is a **stateless firewall** applied at subnet level.

What “stateless” means:

AWS does NOT track connection state.

👉 You must allow:

- inbound traffic

- outbound response separately

Miss one → connection fails.

Key points:

- Works at **subnet level**
- Supports **ALLOW + DENY**
- Rules processed in **number order**
- More control, more mistakes

Then I have installed and started the Apache httpd Web Server.

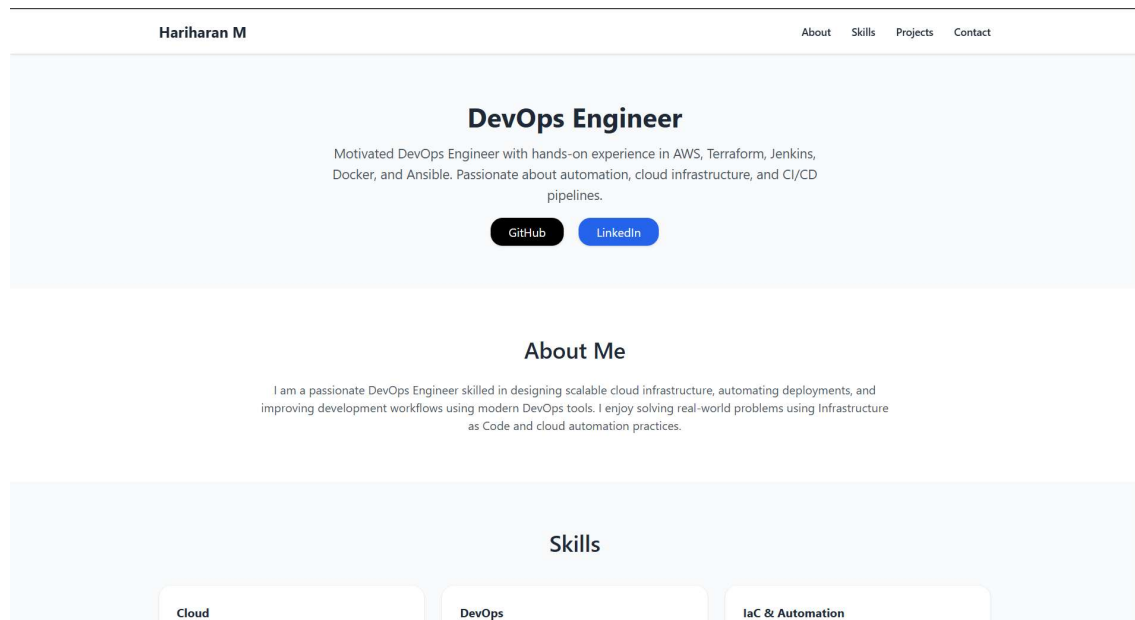
```

[root@ip-10-0-0-58 templatememo.com]# /var/www/html
bash: /var/www/html: is a directory
[root@ip-10-0-0-58 templatememo.com]# cd /var/www/html
[root@ip-10-0-0-58 html]# ls
download
[root@ip-10-0-0-58 html]# cd download/
[root@ip-10-0-0-58 download]# ls
templatememo 559 zay shop
[root@ip-10-0-0-58 download]# cd ..
[root@ip-10-0-0-58 html]# cd ..
[root@ip-10-0-0-58 www]# cd ..
[root@ip-10-0-0-58 var]# cd .
[root@ip-10-0-0-58 var]# cd ..
[root@ip-10-0-0-58 /]# exit
exit
[ec2-user@ip-10-0-0-58 ~]$ sudo su
[root@ip-10-0-0-58 ec2-user]# ls
file1 hari templatememo 574 mexant
[root@ip-10-0-0-58 ec2-user]# nano index.html
[root@ip-10-0-0-58 ec2-user]# mv index.html /var/www/html
[root@ip-10-0-0-58 ec2-user]# cd /var/www/html
[root@ip-10-0-0-58 html]# ls
download index.html
[root@ip-10-0-0-58 html]# systemctl status httpd
● httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; preset: disabled)
   Active: active (running) since Wed 2026-03-18 17:12:23 UTC; 20min ago
     Docs: man:httpd.service(8)
   Main PID: 22501 (httpd)
   Status: "Total requests: 2; Idle/Busy workers 100/0;Requests/sec: 0.00164; Bytes served/sec: 1 B/sec"
   Tasks: 177 (limit: 1067)
   Memory: 13.5M
    CPU: 1.172s
   CGroup: /system.slice/httpd.service

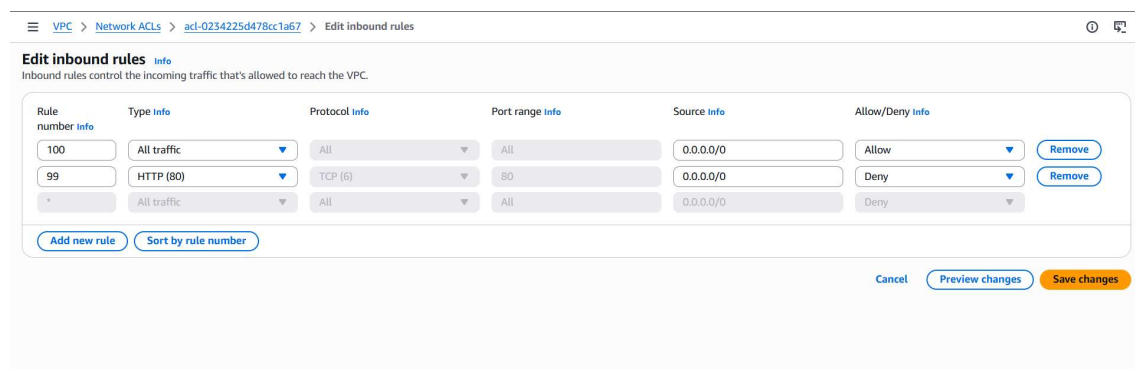
i-0ad7e59fa394f20ca (myec2)
PublicIPs: 5.109.216.136 PrivateIPs: 10.0.0.58

```

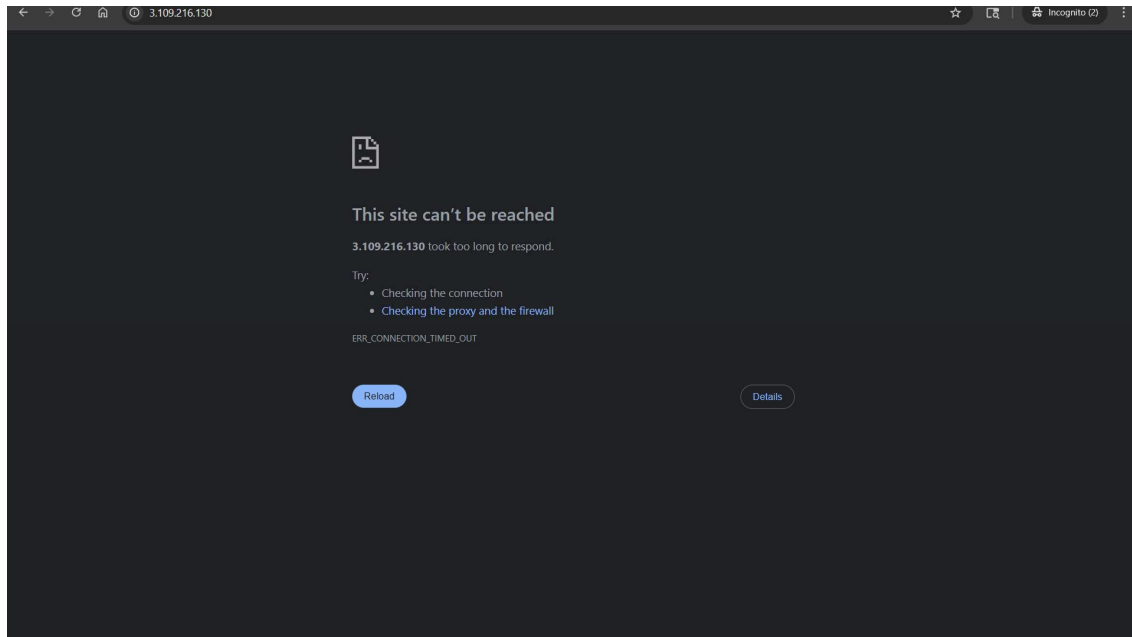
Then I have hosted the http web server my portfolio.



Then using NACL I have denied the http port in the inbound rules.



Then it shows the website can't be reached.



Elastic IP (EIP)

What it is:

An **Elastic IP** is a **static public IPv4 address** in AWS that you own and can attach to resources.

Why it exists:

Normal public IPs in AWS **change when you stop/start EC2** → useless for real systems.

EIP solves that by giving you a **fixed IP**.

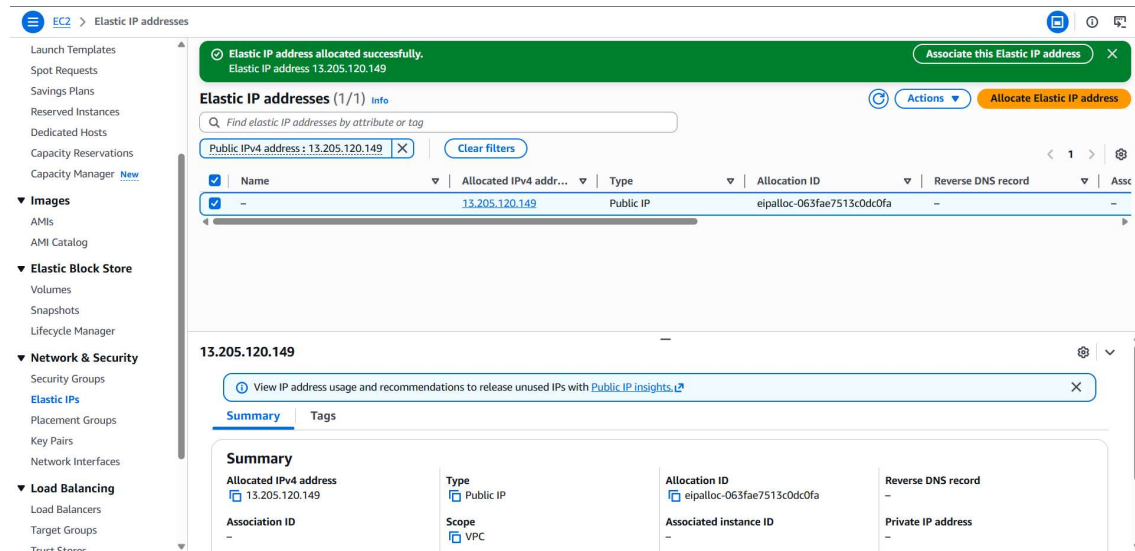
Key points:

- Static (does NOT change)
- You can **attach/detach** from instances
- Works with:
 - EC2
 - NAT Gateway
 - Load Balancer (indirectly)

Real use:

- Hosting a website with a fixed IP
- Whitelisting IP in firewalls
- NAT Gateway outbound internet

Elastic IP Allocation.



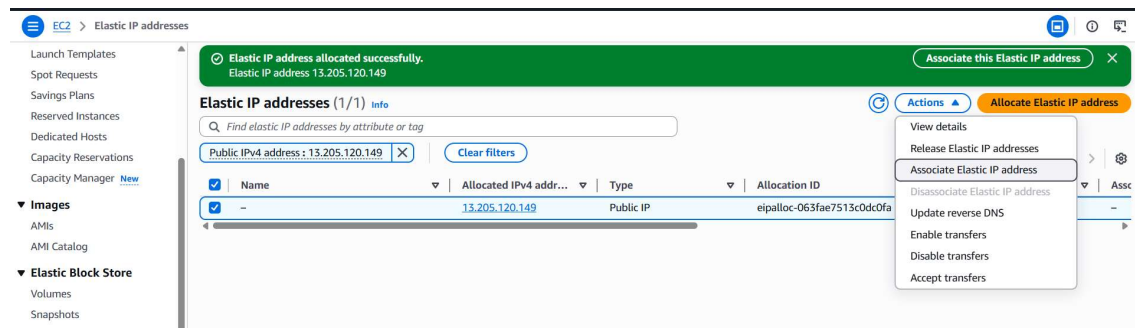
The screenshot shows the AWS Management Console interface for Elastic IP addresses. A green notification banner at the top states "Elastic IP address allocated successfully. Elastic IP address: 13.205.120.149". Below this, the "Elastic IP addresses (1/1)" section displays a table with one entry:

Name	Allocated IPv4 address	Type	Allocation ID	Reverse DNS record	Association ID
-	13.205.120.149	Public IP	eipalloc-063fae7513c0dc0fa	-	-

Below the table, the "Summary" tab is active, showing details for the IP address 13.205.120.149:

- Allocated IPv4 address:** 13.205.120.149
- Type:** Public IP
- Allocation ID:** eipalloc-063fae7513c0dc0fa
- Reverse DNS record:** -
- Association ID:** -
- Scope:** VPC
- Associated instance ID:** -
- Private IP address:** -

After creation we have to associate the Elastic IP.



This screenshot shows the same Elastic IP address page, but with the "Actions" dropdown menu open. The menu options are:

- View details
- Release Elastic IP addresses
- Associate Elastic IP address** (highlighted)
- Disassociate Elastic IP address
- Update reverse DNS
- Enable transfers
- Disable transfers
- Accept transfers

EC2 > Elastic IP addresses > Associate Elastic IP address

Associate Elastic IP address [Info](#)

Choose the instance or network interface to associate to this Elastic IP address (13.205.120.149)

Elastic IP address: 13.205.120.149

Resource type
Choose the type of resource with which to associate the Elastic IP address.

☒ Instance

☐ Network interface

Instance

Private IP address
The private IP address with which to associate the Elastic IP address.

Reassociation
Specify whether the Elastic IP address can be reassociated with a different resource if it already associated with a resource.

☐ Allow this Elastic IP address to be reassociated

[Cancel](#) [Associate](#)

After association of Elastic IP the public IP is Available after Stopage of Instance.

EC2 > Instances > i-001047c49fbc7a5f

EC2

- Dashboard
- AWS Global View
- Events
- Instances**
 - Instances
 - Instance Types
 - Launch Templates
 - Spot Requests
 - Savings Plans
 - Reserved Instances
 - Dedicated Hosts
 - Capacity Reservations
 - Capacity Manager
- Images**
 - AMIs
 - AMI Catalog
- Elastic Block Store**
 - Volumes
 - Snapshots
 - Lifecycle Manager
- Network & Security**

Instance summary for i-001047c49fbc7a5f (myec3) [Info](#)

Updated less than a minute ago

Instance ID i-001047c49fbc7a5f	Public IPv4 address 13.205.120.149 open address	Private IPv4 addresses 172.31.47.192
IPv6 address -	Instance state Running	Public DNS ec2-13-205-120-149.ap-south-1.compute.amazonaws.com open address
Hostname type IP name: ip-172-31-47-192.ap-south-1.compute.internal	Private IP DNS name (IPv4 only) ip-172-31-47-192.ap-south-1.compute.internal	Elastic IP addresses 13.205.120.149 [Public IP]
Answer private resource DNS name IPv4 (A)	Instance type t3.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendation s. Learn more
Auto-assigned IP address -	VPC ID vpc-0f6f0945c28581727	Auto Scaling Group name -
IAM role -	Subnet ID subnet-02b012a063ed00d2e	Managed false
IMDSv2 Required	Instance ARN arn:aws:ec2:ap-south-1:051741041276:instance/i-001047c49fbc7a5f	
Operator -		